Instruction Tuning (Chat-Style)

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Agenda

- SELF-INSTRUCT: Aligning Language Models with (Wang et al., 2023)
- Self-Alignment with Instruction Backtranslation (Li et al., 2024)
- LIMA: Less Is More for Alignment (Zhou et al., 2023)

What is Instruction Tuning?

LMs are pretrained to predict the next token, providing a wide range of knowledge but not making them helpful to users with its outputs.

Instruction tuning finetunes a pretrained model on (instruction, output) pairs, helping the model learn to adhere to user requests and format its response accordingly

- Multi-task Style → tuning on standard NLP tasks to generalize across many different task types.
- Chat Style → tuning on open-ended, user-oriented instructions, making them practical to everyday chat use.

Motivation

The dominant method before 2023 was to collect large, human-annotated instruction datasets (PromptSource, Super-NaturalInstructions aka SuperNI)

Although effective, these sets were difficult to build

- Costly → requiring experts significant amounts of work to make large sets
- Limited Diversity → limited by the creativity of the human experts

The following papers each challenge this assumed dataset paradigm, finding ways to reduce dependence on large-scale human annotation while maintaining high model performance.

SELF-INSTRUCT: Aligning Language Models with Self-Generated Instructions

Yizhong Wang♣ Yeganeh Kordi♢ Swaroop Mishra♡ Alisa Liu♣ Noah A. Smith♣+
Daniel Khashabi♠ Hannaneh Hajishirzi♣+

♣University of Washington ♢Tehran Polytechnic ♡Arizona State University
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Introduction

- Motivation
- Overview

Motivation

Language Models (LMs) are great at generalizing zero-shot to new tasks.

This requires instruction tuning, which depends heavily on human-written instruction data.

- Often limited in quantity, creativity, diversity.
- Often expensive or difficult to obtain

Overview

Self-Instruct introduces the idea of taking a small, high quality human-annotated dataset and using a pretrained, off-the-shelf LM to generate new instructions with input-output pairs using iterative bootstrapping.

Core contributions of the paper

1) Self-Instruct method of instruction fine-tuning with minimal human data
2) Demonstrate the effectiveness of self-instruct
3) Release large synthetic dataset of 52,000 instructions

Method

- Defining Instruction Data
- Automatic Instruction Data Generation
- Finetuning the LM to Follow Instructions

Method

Defining Instruction Data

$\{I_t\}$ → A set of instructions

Each defines a task "*t*" in natural language

Task *t* has $n_t \geq 1$ input-output instances

$\{(X_{t,i}, Y_{t,i})\}$ for $i = 1 \ldots n_t$

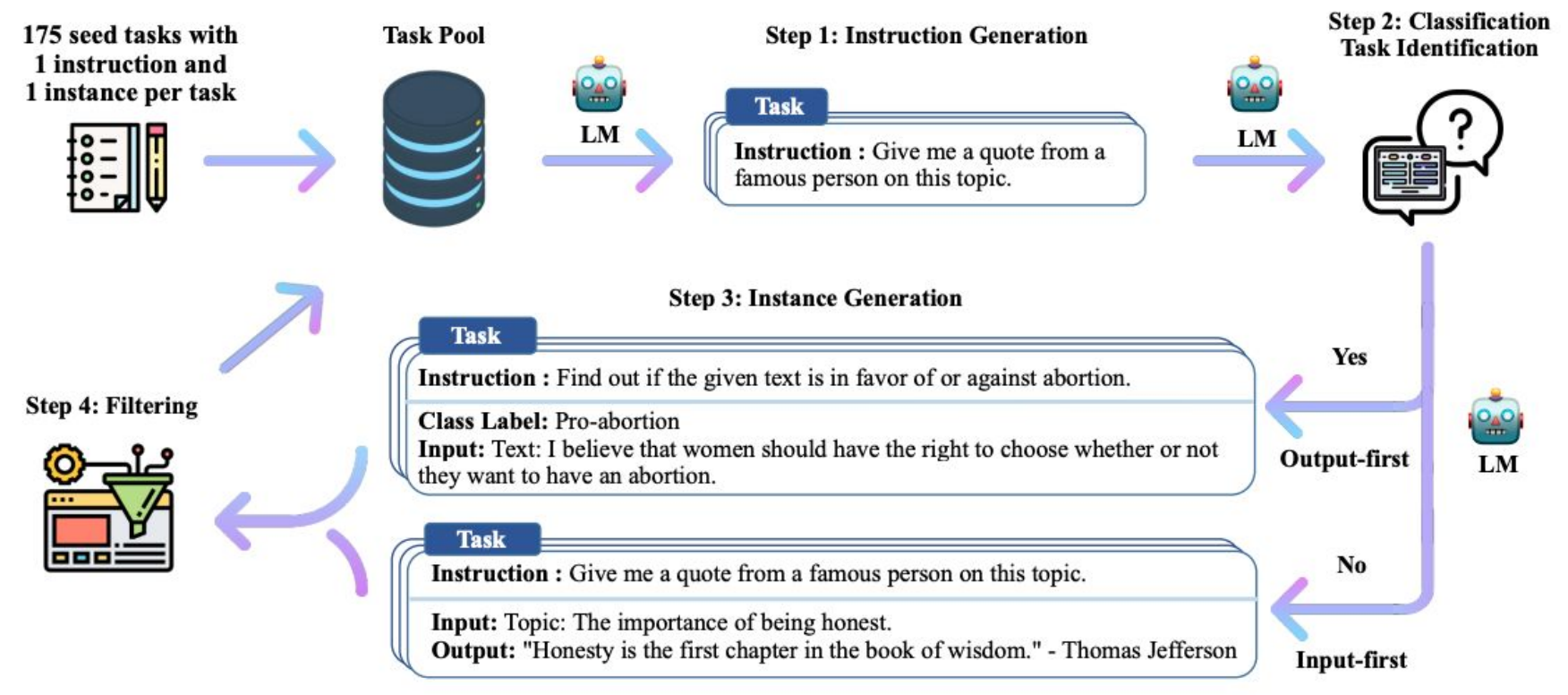
NB: input can be null, tasks can have multiple instances

A model is expected to produce the output given the task and input

$M(I_t, X_{t,i}) = Y_{t,i}$, for $i \in \{1, \ldots, n_t\}$

Method

Automatic Instruction Data Generation



Method

Automatic Instruction Data Generation

1) <u>Generating Task Instructions</u>
   - Initialize the task pool with 175 tasks (1 instance per task).
   - For each sample, draw 8 (6 human written, 2 machine written).
   - Generate new instruction.

2) <u>Determining classification tasks</u>
   - Use few-shot prompting with the LM to determine if an instruction is a classification task
   - *Crucial* step because LMs that generate input first are often biased towards labels in classification tasks

Method

Automatic Instruction Data Generation

3) <u>Instance Generation</u>

- *Input-First* → generate the input, and then the output (standard)
- *Output-First* → generate the output, then the input (classification tasks)

```
Given the classification task definition and the class labels, generate an input that
corresponds to each of the class labels.  If the task doesn't require input, just generate the
correct class label.

Task:  Classify the sentiment of the sentence into positive, negative, or mixed.
Class label:  mixed
Sentence:  I enjoy the flavor of the restaurant but their service is too slow.
Class label:  Positive
Sentence:  I had a great day today.  The weather was beautiful and I spent time with friends.
Class label:  Negative
Sentence:  I was really disappointed by the latest superhero movie.  I would not recommend it.
```

Method

Automatic Instruction Data Generation

4) <u>Filtering</u>
   - For diversity, only add a task that has a ROUGE-L similarity score less than .7 with existing instruction pool (standard metric ranging from 0-1, lower scores mean no similarity)
   - Drop repeat instances
   - Drop invalid generations
     - Heuristic recognized
     - Ex: output instance is too short or too long

Method

Finetuning the LM to Follow Instructions

After generating all instructions, fine tune the original LM

Concatenate instruction and its input-output instances into a prompt, and then proceed to instruction tune in the standard supervised way

Incorporate different prompt formats for robustness

- Include random line break, include mixes of labeled and unlabeled prompts
  - “Instruction: <instruction> Input: <input>”
  - “<instruction> <input>”
  - “Task: <instruction> <input>”

Self-Instruct Data From GPT3

- Statistics
- Diversity
- Quality

Self-Instruct Data From GPT3

Statistics

Case study performed using GPT3 as the vanilla LM.

Instruction tuned LM:

$GPT3_{SELF\text{-}INST}$

Generated *52K tasks* with *82K* instances

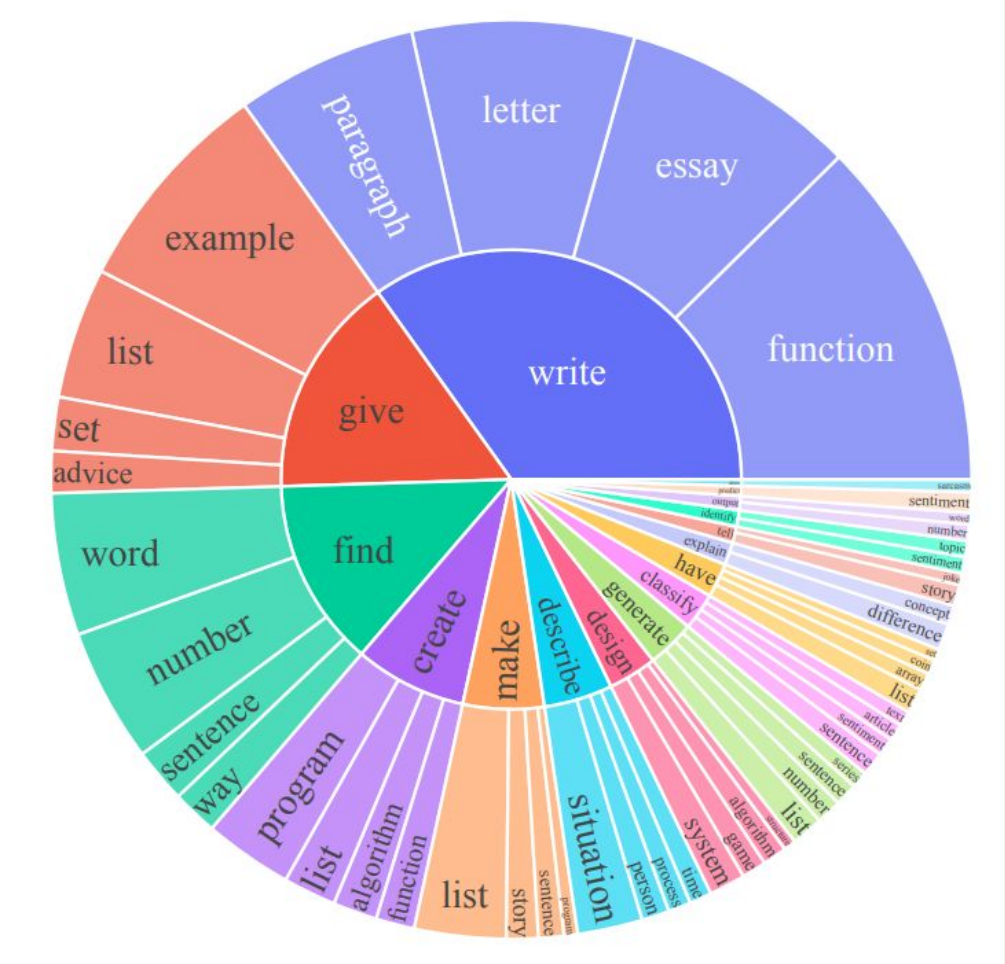
# of instructions	52,445
- # of classification instructions	11,584
- # of non-classification instructions	40,861
# of instances	82,439
- # of instances with empty input	35,878
ave. instruction length (in words)	15.9
ave. non-empty input length (in words)	12.7
ave. output length (in words)	18.9

Self-Instruct Data From GPT3

Diversity

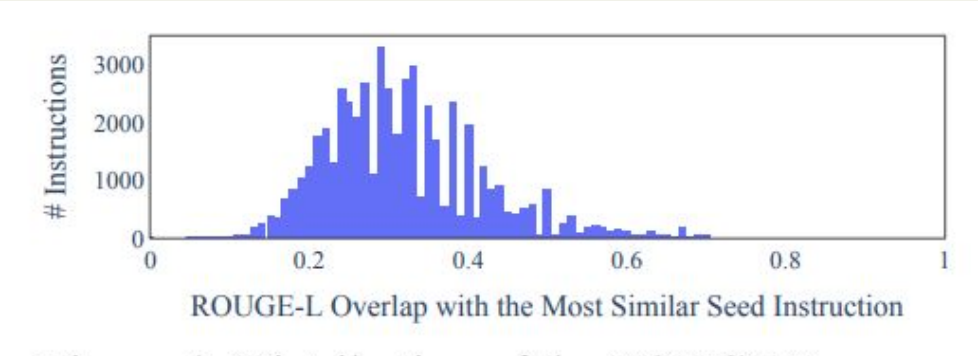
Used the Berkeley Neural Parser

- ~50% of the data contained a root verb and direct noun
  - Remaining data had more complicated structure
- Here, displaying top 20 verbs and their top 4 direct nouns
  - Only 14% of the total data

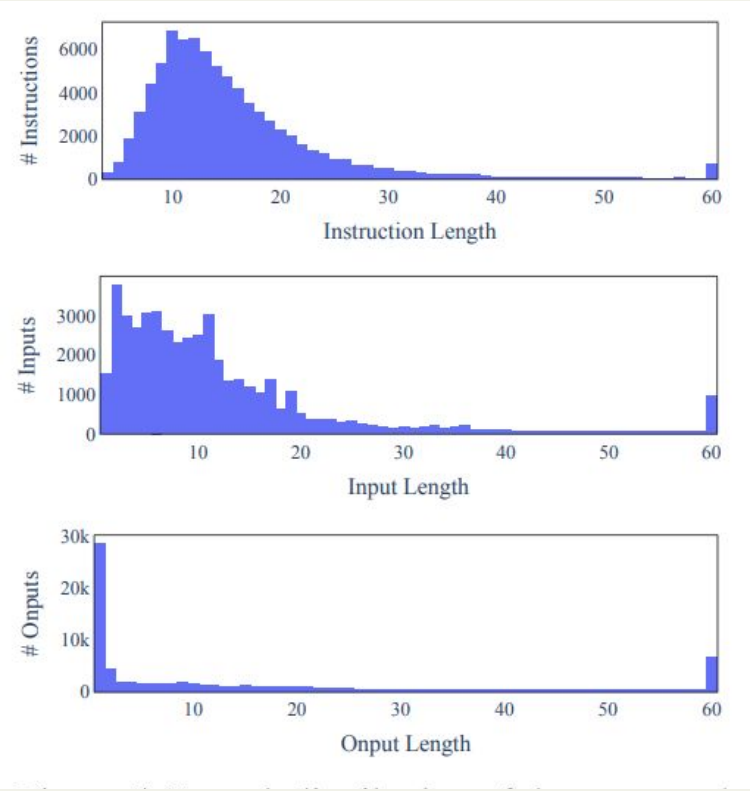


Self-Instruct Data From GPT3

Diversity



Distribution of highest ROUGE-L scores of generated data and seed data



General diversity metrics

Self-Instruct Data From GPT3

Quality

Randomly sample 200 instructions (1 instance per)

Experts labeled whether each instance is correct or not, in terms of the instruction, the input, and the output

Quality Review Question	Yes %
Does the instruction describe a valid task?	92%
Is the input appropriate for the instruction?	79%
Is the output a correct and acceptable response to the instruction and input?	58%
All fields are valid	54%

Instructions were very good

Instances were ok (noisy but reasonable)

Experimental Results

- Experiments Overview
- Experiment 1: Zero-Shot Generalization on SUPERNI Benchmark
- Experiment 2: Generalization to User-Oriented Instruction On Novel Tasks
- Effects of Data Size and Quantity

Experiments Overview

Testing Self-Instruct from two different stances

Both zero-shot prompting against vanilla and instruction tuned models

Uses GPT3 as the base for Self-Instruct → $GPT3_{Self\text{-}Inst}$

1) Traditional NLP Tasks

2) User-Oriented NLP Tasks

Experiment 1: Traditional NLP Tasks

Benchmark → unseen SuperNI tasks

119 tasks (subset of SuperNI) , zero shot, ROUGE-L metric

<u>Vanilla</u>

- T5-LM
- GPT3

<u>Instruction Tuned (w/o SuperNI)</u>

- T0 (PromptSource)
- GPT3 + T0 (PromptSource)
- $GPT3_{Self\text{-}Inst}$
- InstructGPT_{001} (private)

<u>Instruction Tuned (w/ SuperNI)</u>

- T*k*-Instruct (SuperNI)
- GPT3 + SuperNI
- $GPT3_{Self\text{-}Inst}$ *+ SuperNI*

InstructGPT_{001} chosen as newer models likely have already seen SuperNI

Experiment 1: Traditional NLP Tasks

- $GPT_{Self\text{-}Inst}$ improves upon the base model by 33.1%
- $GPT_{Self\text{-}Inst}$ scores higher than T0 or GPT3 with T0 training
- $GPT_{Self\text{-}Inst}$ nearly matches $InstructGPT_{001}$
- When combined with SuperNI training, $GPT_{Self\text{-}Inst}$ still improves

	Model	# Params	ROUGE-L
	Vanilla LMs		
	T5-LM	11B	25.7
	GPT3	175B	6.8
	Instruction-tuned w/o SuperNI		
1	T0	11B	33.1
	GPT3 + T0 Training	175B	37.9
2	$GPT3_{Self\text{-}Inst}$ (Ours)	175B	39.9
	$InstructGPT_{001}$	175B	**40.8**
	Instruction-tuned w/ SuperNI		
	T*k*-Instruct	11B	46.0
3	GPT3 + SuperNI Training	175B	49.5
	$GPT3_{Self\text{-}Inst}$ + SuperNI Training (Ours)	175B	**51.6**

Experiment 2: User-Oriented Tasks

Benchmark → Custom

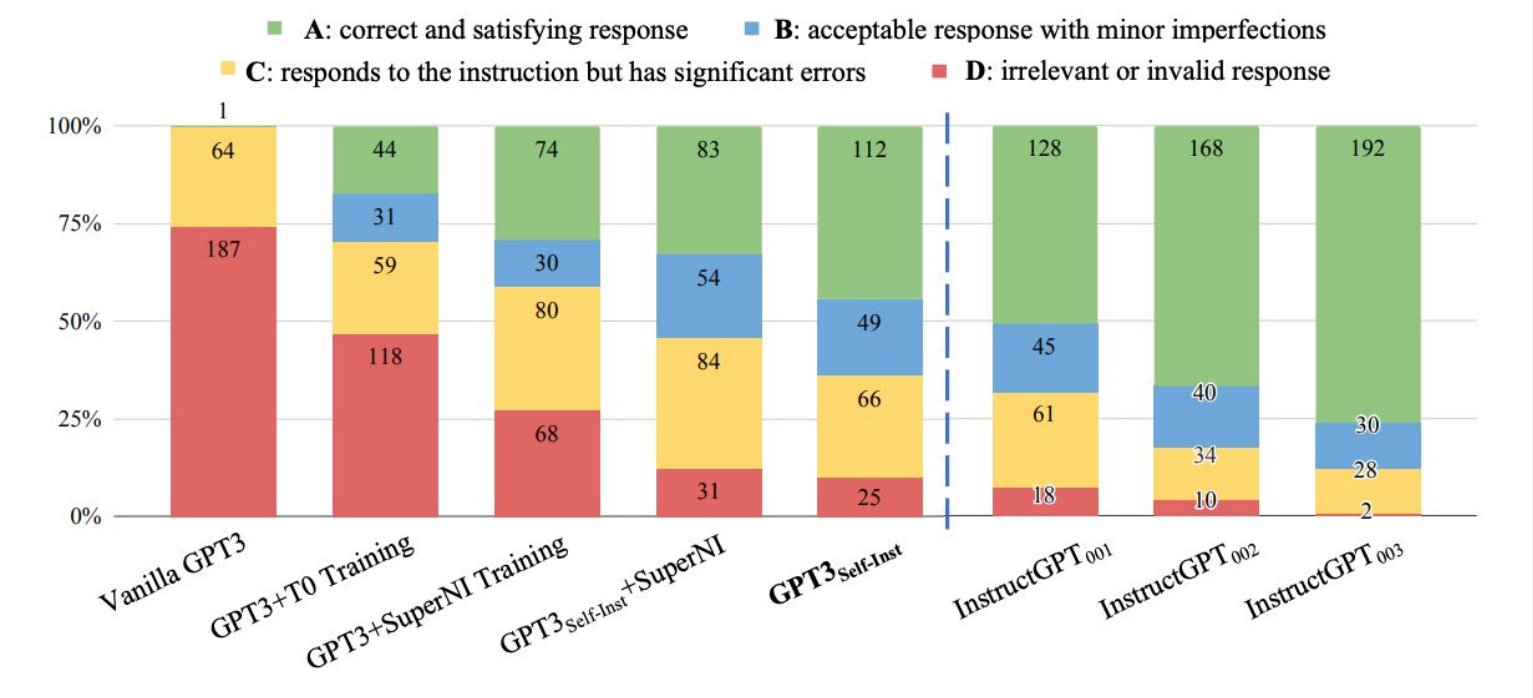
- Built by considering user domains → more practical than SuperNI, which is built for research and focuses around classification
- 252 instructions (1 per instance)
- Diverse, novel, and difficult to quantity tasks (Authors also act as judges)
    - Rating system from A-D

Models

- Uses previous GPT3 models
- Additional newer InstructGPT_{002} and InstructGPT_{003} as the test set is completely novel.

Experimental Results

Experiment 2: User-Oriented Tasks



$GPT3_{Self\text{-}Inst}$ 5% behind $InstructGPT_{001}$

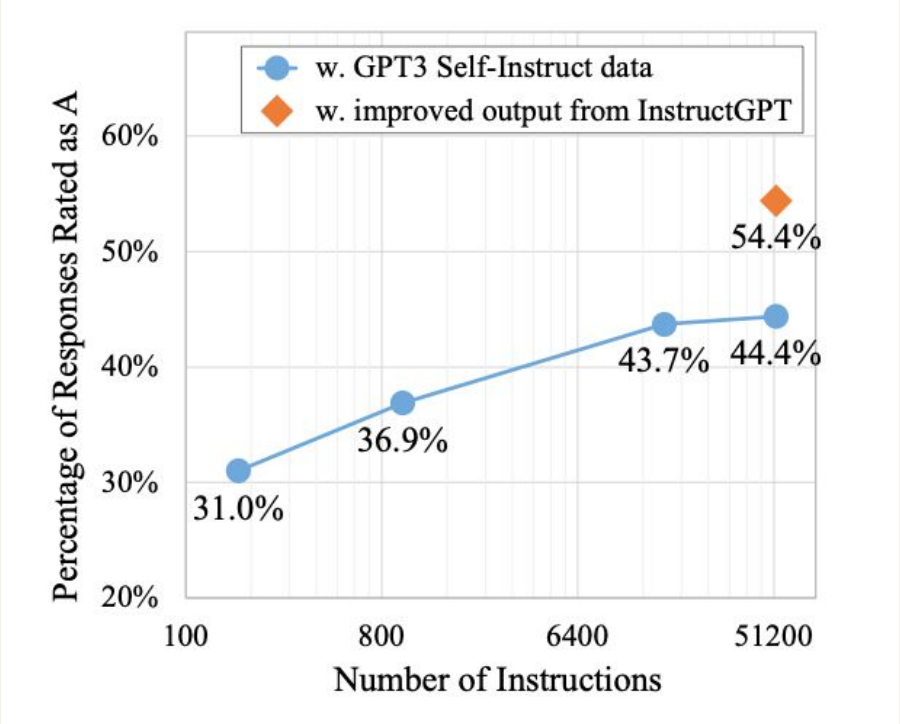
Experimental Results

Effects of Data Size and Quantity

Repeat experiment on User-Oriented Tasks

- Focusing on $GPT_{Self\text{-}Inst}$ and changing its training size from the synthetic data
- <u>Trend</u> → more data leads to better results (plateaus around 16k)

Additionally, redid it with $InstructGPT_{003}$ re-generating the outputs in the dataset→ large improvement (10%)



Broader Impacts + Importance

Introduced a new way to collect fine-tuning data with minimal human work.

Transparency in instruction-tuned models

- Models like InstructGPT are hidden → difficult to understand the reason for its capabilities
- Encouraging future transparency and emphasising its importance.

Demonstrates the importance of diverse instruction data

- The large, synthetic data resulting from GPT3 is a first step towards higher-quality public data for building better instruction-following models.

Limitations

Tail Phenomena

- *What it is* → LMs see far more head examples during pretraining, so they are fundamentally stronger at common things than rare ones.
- *Effect* → (1) generation will gravitate towards common tasks and (2) the LM is likely to struggle with uncommon tasks.

Dependence on Large Models

- Likely requires large models for quality data generation, creating a barrier of access

Reinforcing LM Biases

- Models prior biases can be reflected in generated data.

Self-Alignment with Instruction Backtranslation

Xian Li, Ping Yu, Chunting Zhou, Timo Schick, Omer Levy, Luke Zettlemoyer
Jason Weston & Mike Lewis

Meta

Introduction

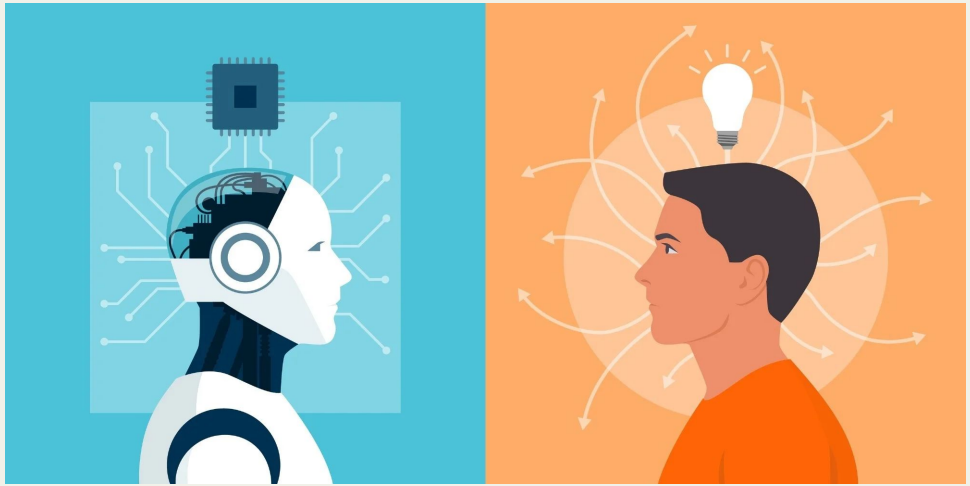
- Motivation
- Overview

Motivation

Same motivation as for Self-Instruct

Instruction tuning is important for model performance but heavily dependant on human-written instruction data

- Expensive
- Limited in quantity, creativity, diversity



Overview

Scalable method for improving LLMs' instruction following by generating instruction tuning training data from large amounts of unlabeled text:

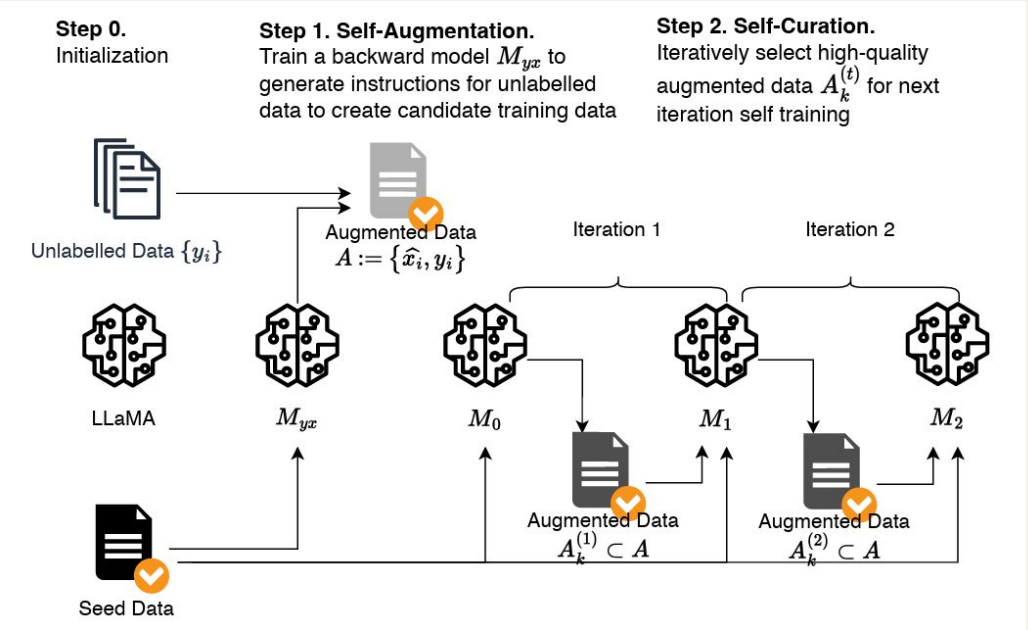
Backtranslation

1. The model generates candidate instructions for web document snippets (self-augmentation)
2. Filters high quality instruction & response pairs to iteratively improve itself (self-curation)

Outperforms all other existing non-distilled models on the Alpaca leaderboard

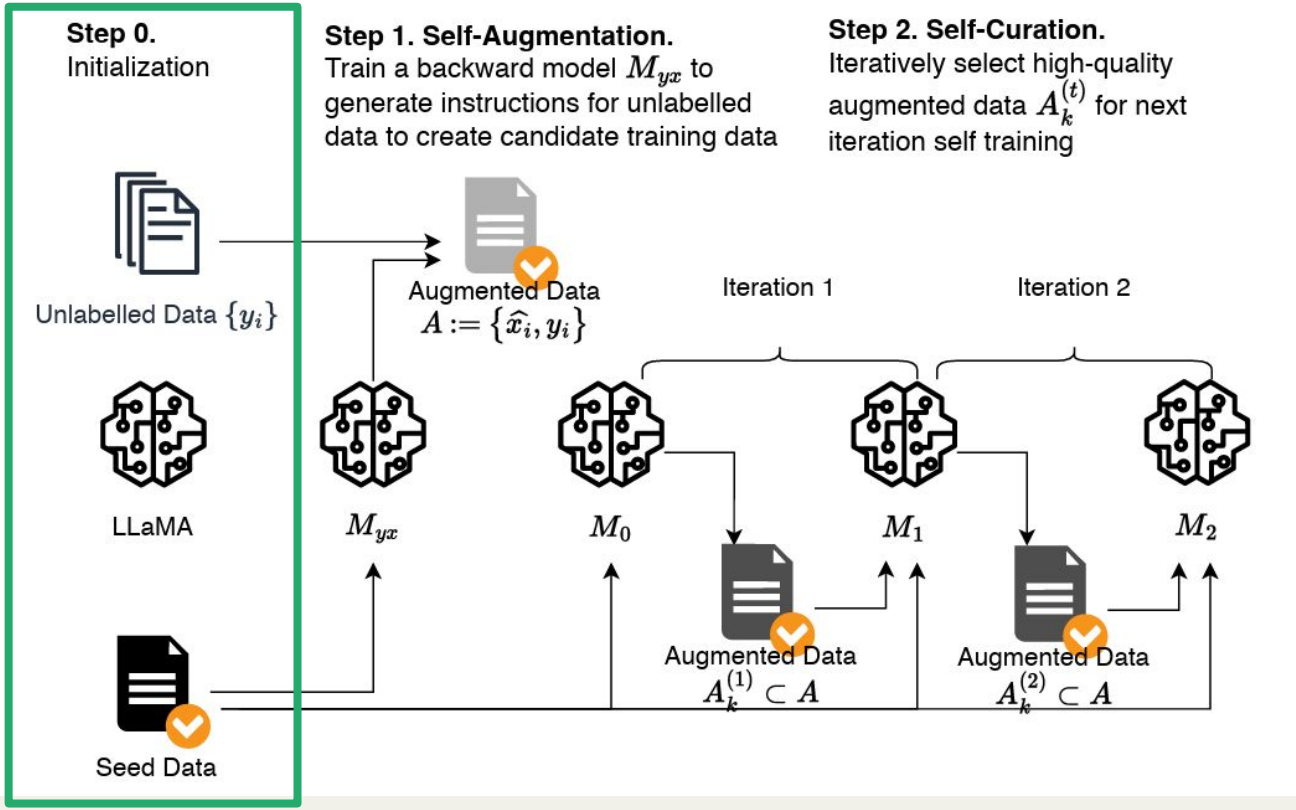
Method

- Backtranslation



Method

Backtranslation



Step 0: Initialization

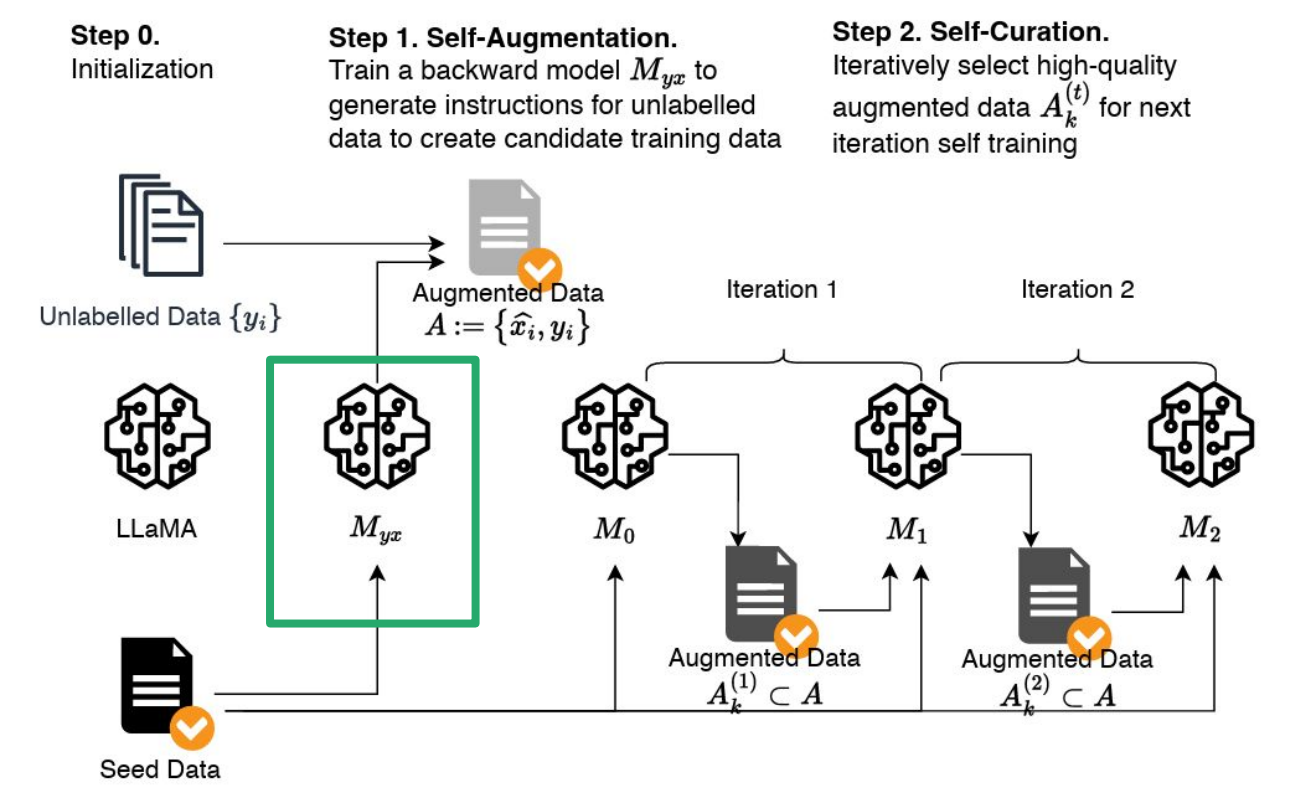
Unlabelled Data: preprocessed web corpus (Clueweb)

Base Model: pretrained LLaMA w/ 7B, 33B and 65B parameters

Seed Data: human-annotated (instruction, output) pairs

Method

Backtranslation

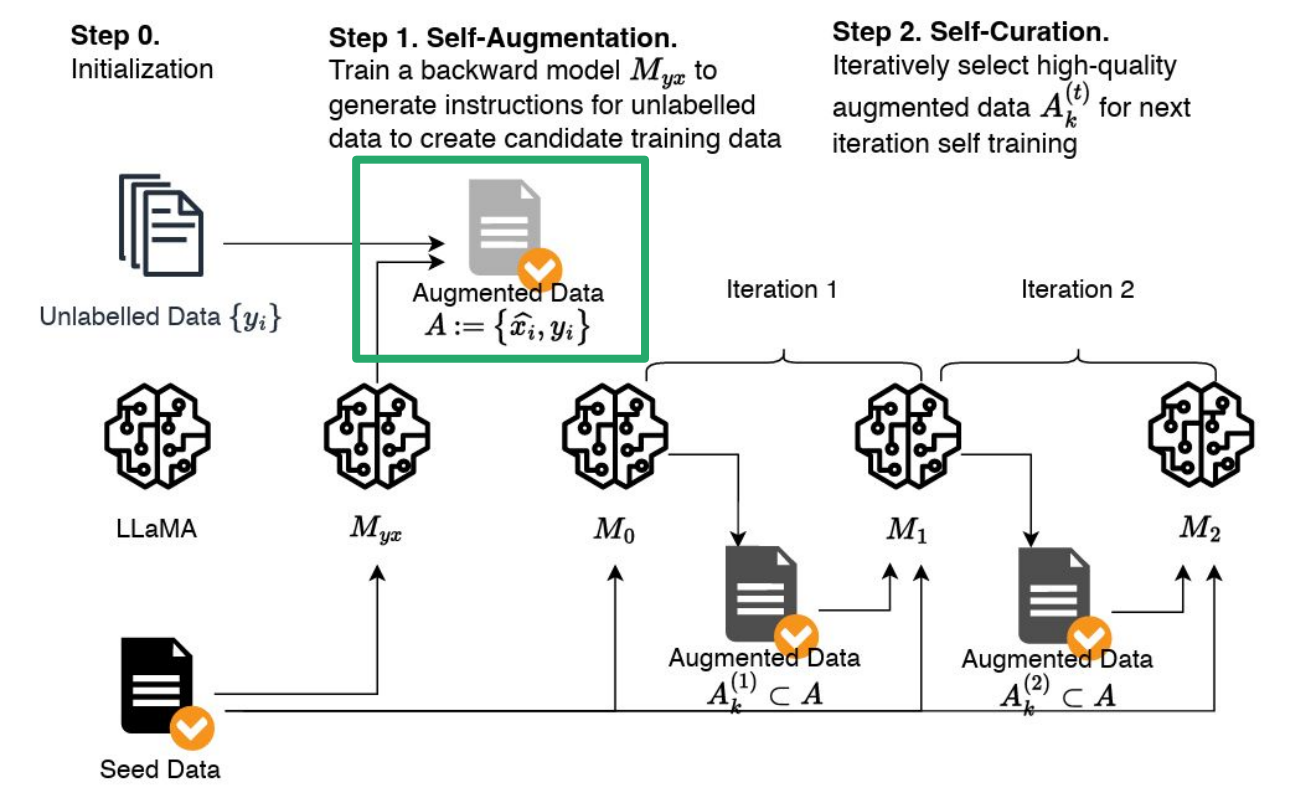


Step 1: Self-Augmentation

Backward Model: finetune the base model w/ Seed Data $\{(y_i,x_i)\}$ to obtain a backward model $M_{yx} := p(x|y)$

Method

Backtranslation



Step 1: Self-Augmentation

Augmented Data: For each unlabelled example y_i, run inference on the backward model to generate a candidate instruction $\hat{x}_i$ to create pairs $A := \{(\hat{x}_i, y_i)\}$

*Note: Quality varies

Method

Backtranslation

Output:

Diamond engagement rings gained in popularity during the Art Deco era with the round old European cut diamond being the favourite.

Asscher Cut

The Asscher cut is one of the first patented diamond cuts in the world and was invented by Dutch master diamond cutter, Joseph Asscher of the Royal Asscher Diamond Company in 1902. Classic asscher cut diamonds are cut into squares and resemble emerald cuts, which are rectangular. Asscher cut diamonds are different to a square emerald cut in that they have larger step facets, a higher crown, smaller table and have more brilliance. The corners are cropped to give the shape an octagonal appearance.

Baguette Cut

Although the baguette cut was invented sometime prior to the mid-1500s, it only gained popularity in 1912 when Cartier reintroduced the cut to the modern world. Its elongated, table cut, rectangular shape became highly fashionable in the geometric craze of the Art Deco period.

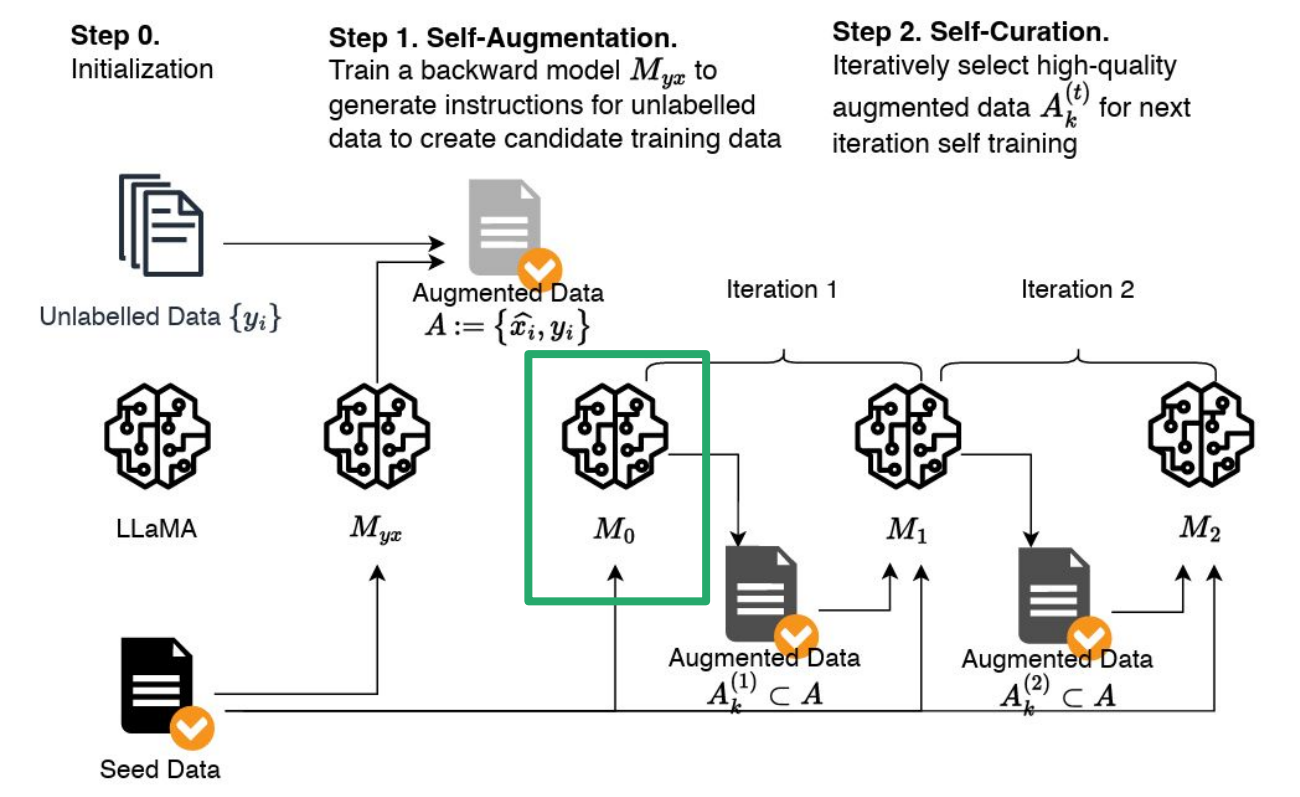
Emerald Cut

The emerald diamond cut emerged as one of the first faceted diamond cuts, third in line after the point cut and the table cut. The cut has a dramatic hall of mirrors effect and was standardised in the 1940s.

Generated instruction: List the most popular diamond cuts in the Art Deco era.

Method

Backtranslation

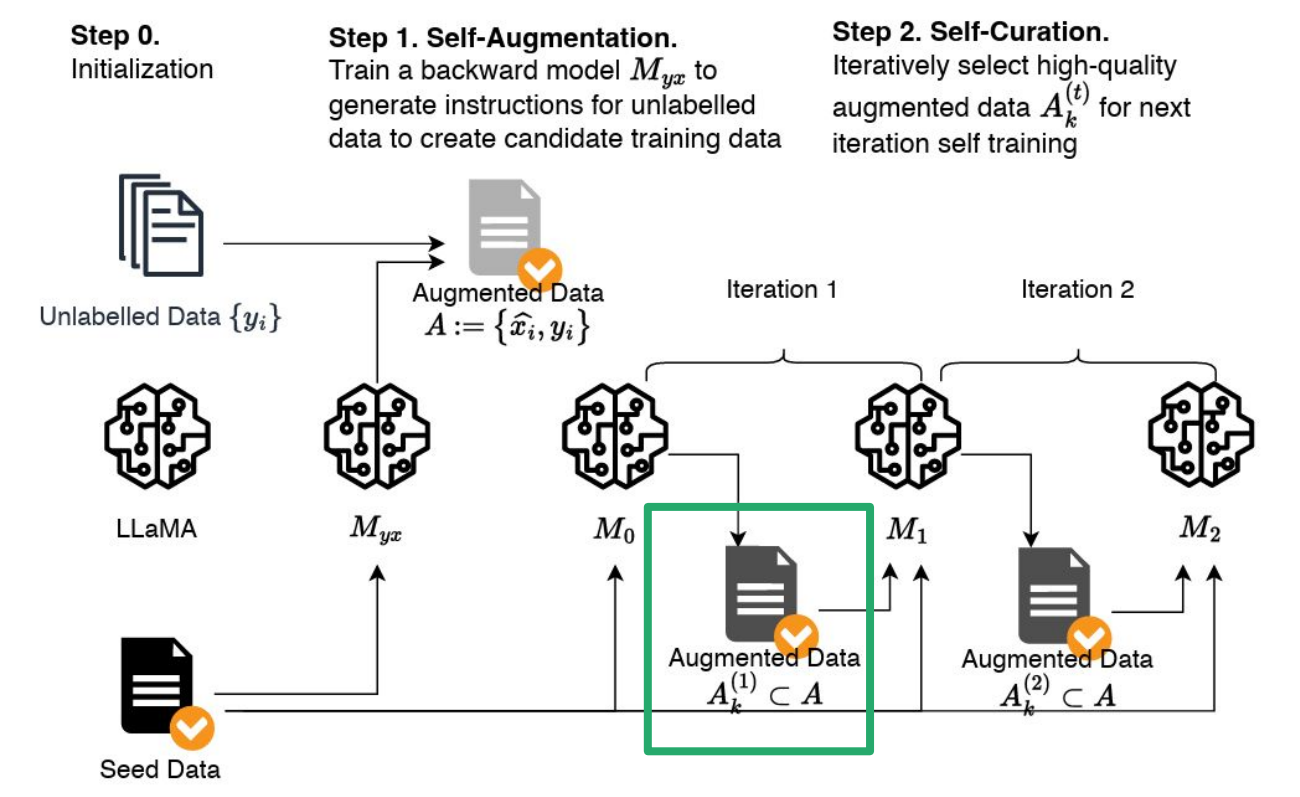


Step 2: Self-Curation

Humpback Model: similar to backward model, initializes by training on Seed Data (M_0)

Method

Backtranslation



Step 2: Self-Curation

Quality Assignment: M_0 scores Augmented Data on 5 point Likert Scale

Curated Set: select subset of the augmented examples with ***score*** $\geq$ k to form $A_k^{(1)}$

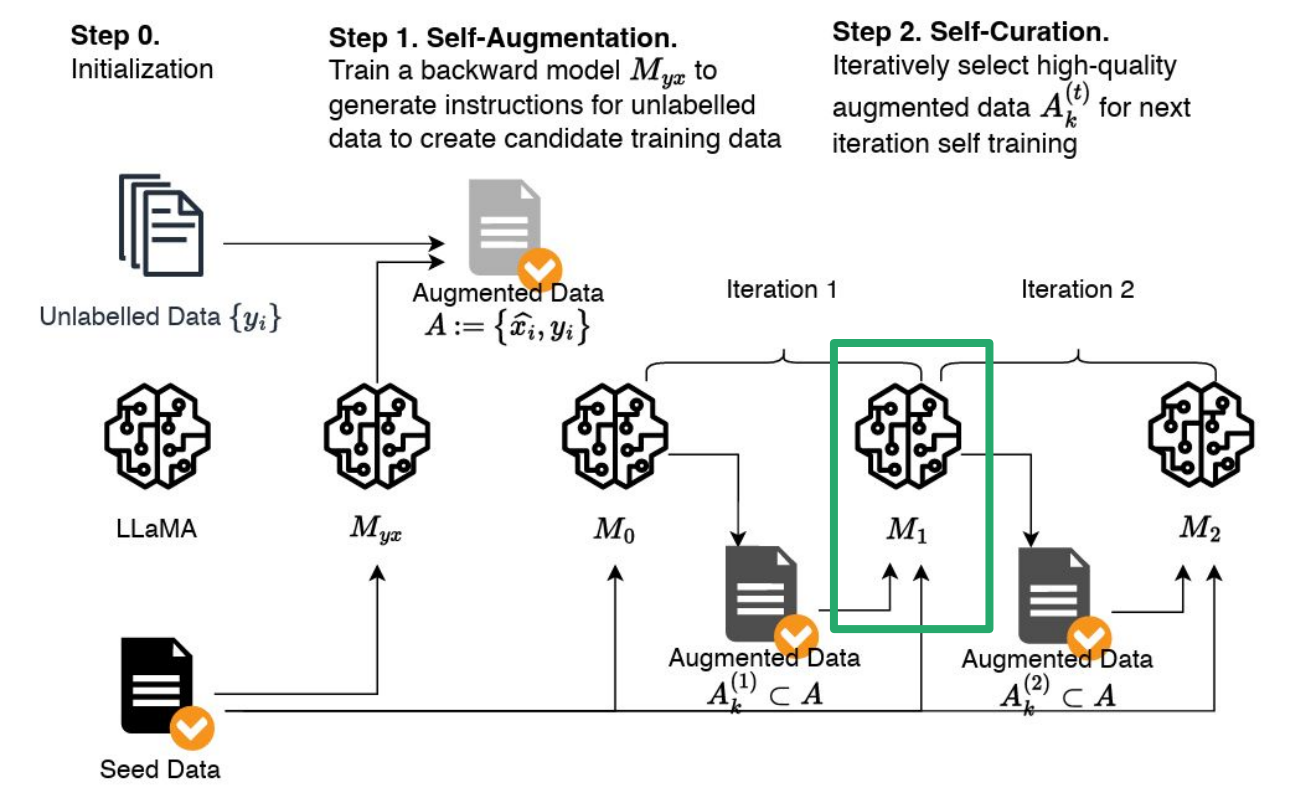
Below is an instruction from an user and a candidate answer. Evaluate whether or not the answer is a good example of how AI Assistant should respond to the user's instruction. Please assign a score using the following 5-point scale:
1: It means the answer is incomplete, vague, off-topic, controversial, or not exactly what the user asked for. For example, some content seems missing, numbered list does not start from the beginning, the opening sentence repeats user's question. Or the response is from another person's perspective with their personal experience (e.g. taken from blog posts), or looks like an answer from a forum. Or it contains promotional text, navigation text, or other irrelevant information.
2: It means the answer addresses most of the asks from the user. It does not directly address the user's question. For example, it only provides a high-level methodology instead of the exact solution to user's question.
3: It means the answer is helpful but not written by an AI Assistant. It addresses all the basic asks from the user. It is complete and self contained with the drawback that the response is not written from an AI assistant's perspective, but from other people's perspective. The content looks like an excerpt from a blog post, web page, or web search results. For example, it contains personal experience or opinion, mentions comments section, or share on social media, etc.
4: It means the answer is written from an AI assistant's perspective with a clear focus of addressing the instruction. It provide a complete, clear, and comprehensive response to user's question or instruction without missing or irrelevant information. It is well organized, self-contained, and written in a helpful tone. It has minor room for improvement, e.g. more concise and focused.
5: It means it is a perfect answer from an AI Assistant. It has a clear focus on being a helpful AI Assistant, where the response looks like intentionally written to address the user's question or instruction without any irrelevant sentences. The answer provides high quality content, demonstrating expert knowledge in the area, is very well written, logical, easy-to-follow, engaging and insightful.

Please first provide a brief reasoning you used to derive the rating score, and then write "Score: <rating>" in the last line.

<generated instruction>
<output>

Method

Backtranslation



Step 2: Self-Curation

Iterative Training: M_0 is further trained on the curated augmentation training data $A_k^{(1)}$ to finetune an improved model M_1. Process iteratively repeats

Data Generated By Backtranslation

- Statistics
- Diversity

Data Generated By Backtranslation

Statistics

	# examples	Instruction Length	Output Length
Seed data	3200	148 ± 322	1072 ± 818
Augmented data, $\mathcal{A}_5^{(2)}$	41821	115 ± 175	1663 ± 616
Augmented data, $\mathcal{A}_4^{(2)}$	195043	206 ± 298	1985 ± 649
Augmented data, all	502133	352 ± 134	1722 ± 653

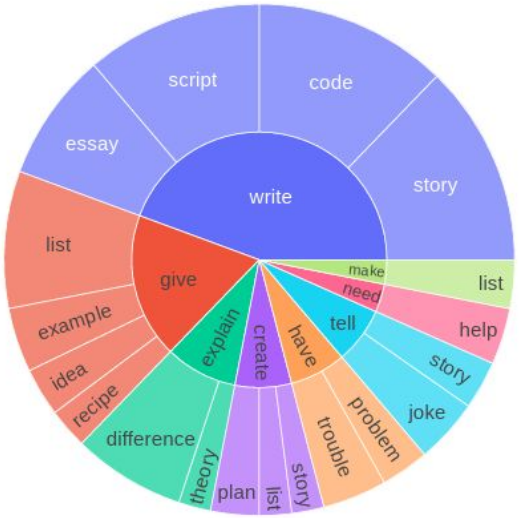
Augmented data generated ~500k total instruction-output instances

- Tends to have longer outputs compared to the seed data
  - Self-curated higher quality training data ($A_4^{(2)}$ and $A_5^{(2)}$) has both shorter instructions and outputs among all augmented data, closer to the length of the original seed instruction data

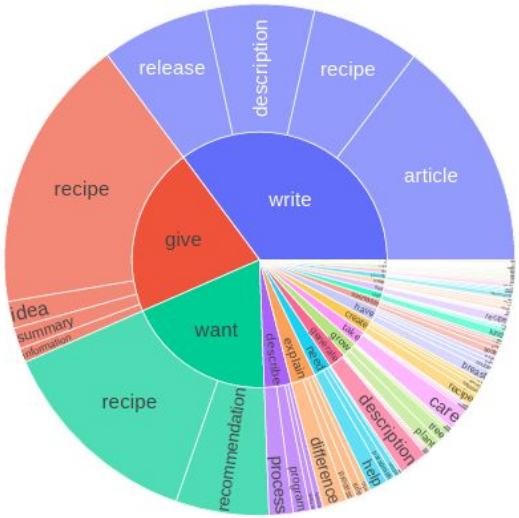
Data Generated By Backtranslation

Diversity

Augmentation data appears to possess diversity especially in the long tail, and to be complementary to the existing human-annotated seed data



(a) Seed data.



(b) Augmented data in $\mathcal{A}_5$

Experiments

- Experiment Setup
- Experiment Results
- Scaling Analysis
- Ablation Studies

Experiment Setup: Humpback

Seed Data: 3200 human-annotated examples from Open Assistant dataset

Unlabelled Data: 502k English segments from Clueweb corpus

Base Model & Finetuning:

- pretrained LLaMA model w/ 7B, 33B and 65B parameters as base model
- only optimize the loss on the output tokens, not the input tokens, during finetuning

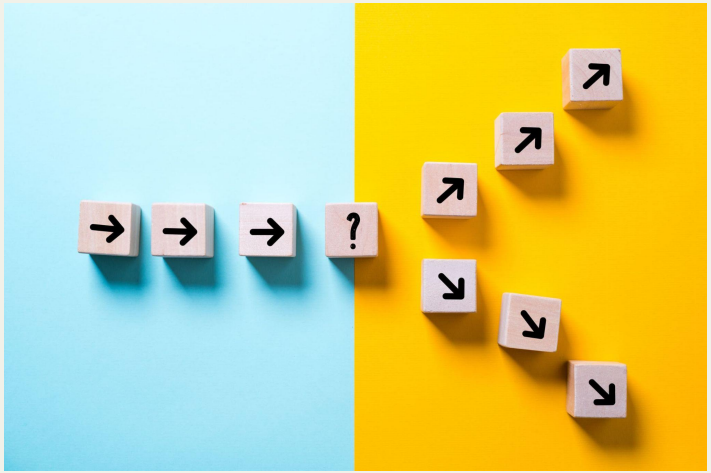
Experiment Setup: Baselines

Direct Comparison Baselines (Non-distilled):

- Text-davinci-003: GPT-3 finetuned with human-written instructions/outputs, model responses and human preferences using RL
- LIMA: LLaMA finetuned with small set of manually selected instruction examples from community Q&A (StackOverflow, WikiHow, etc.) and human expert-written instruction and responses
- Guanaco: LLaMA finetuned with 9000 examples from all turns in OpenAssistant dataset (vs 3200 first turn examples that Humpback uses)

Experiment Setup: Evaluation

- Sample 256 prompts from a mixed set of 1130 prompts covering a variety of task categories
- Output Evaluation:
  - AlpacaEval: GPT-4 judgement to compute win rate against baseline models
  - Human preference evaluation



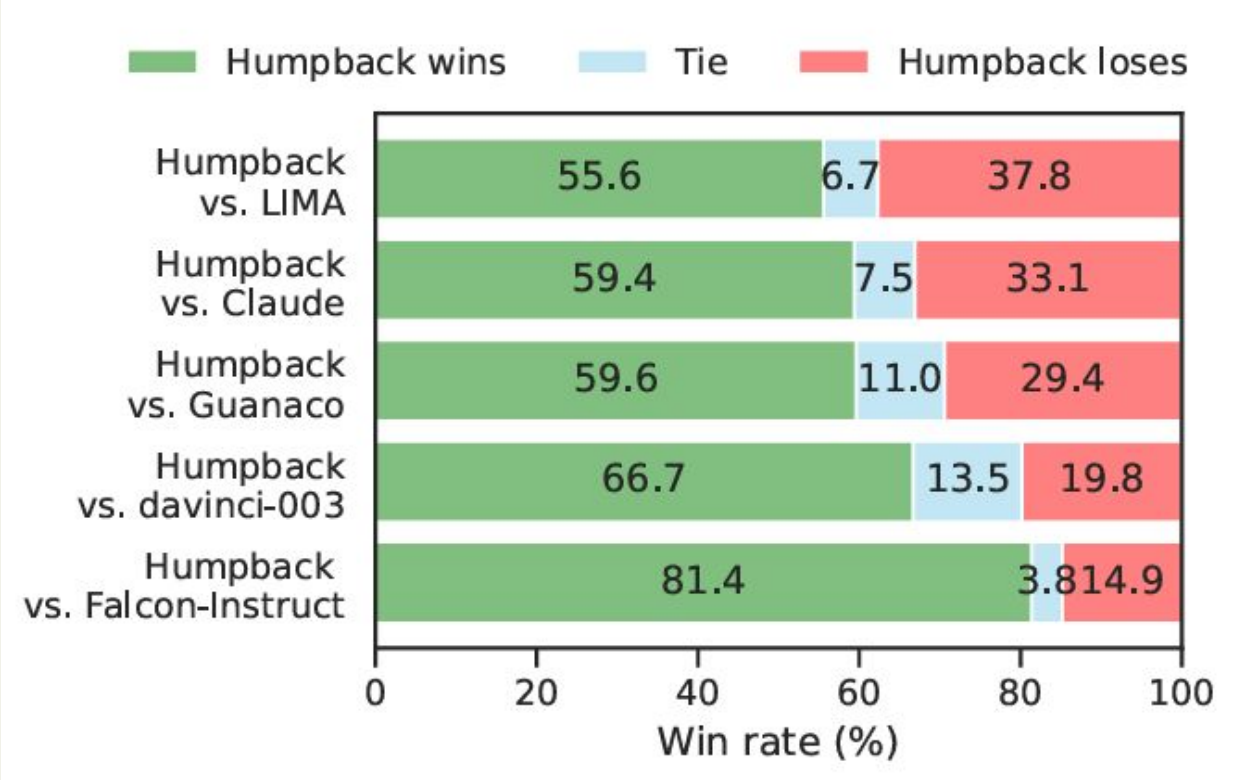
Experiment Results: AlpacaEval

Win rate over text-davinci-003 evaluated by GPT-4

		Annotated Examples	Total Examples	Win Rate %
Non-distilled	Humpback 33B	3k	45k	**79.84**
	OASST RLHF 33B	161k	161k	66.52
	Guanaco 33B	9k	9k	65.96
	OASST SFT 33B	161k	161k	54.97
Non-distilled	Humpback 65B	3k	45k	**83.71**
	Guanaco 65B	9k	9k	71.80
	LIMA 65B	1k	1k	62.70
Non-distilled	Humpback 70B	3k	45k	87.94
	LLaMa2 Chat 70B	1.4m	5.7m	**92.66**
Distilled	Vicuna 33B	140k	140k	**88.99**
	WizardLLM 13B	190k	190k	86.32
	airoboros 65B	17k	17k	73.91
	Falcon Instruct 40B	100k	100k	45.71
Proprietary	GPT-4			**95.28**
	Claude 2			91.36
	ChatGPT			89.37
	Claude			88.39

Experiment Results: Human

Pairwise human preference judgements

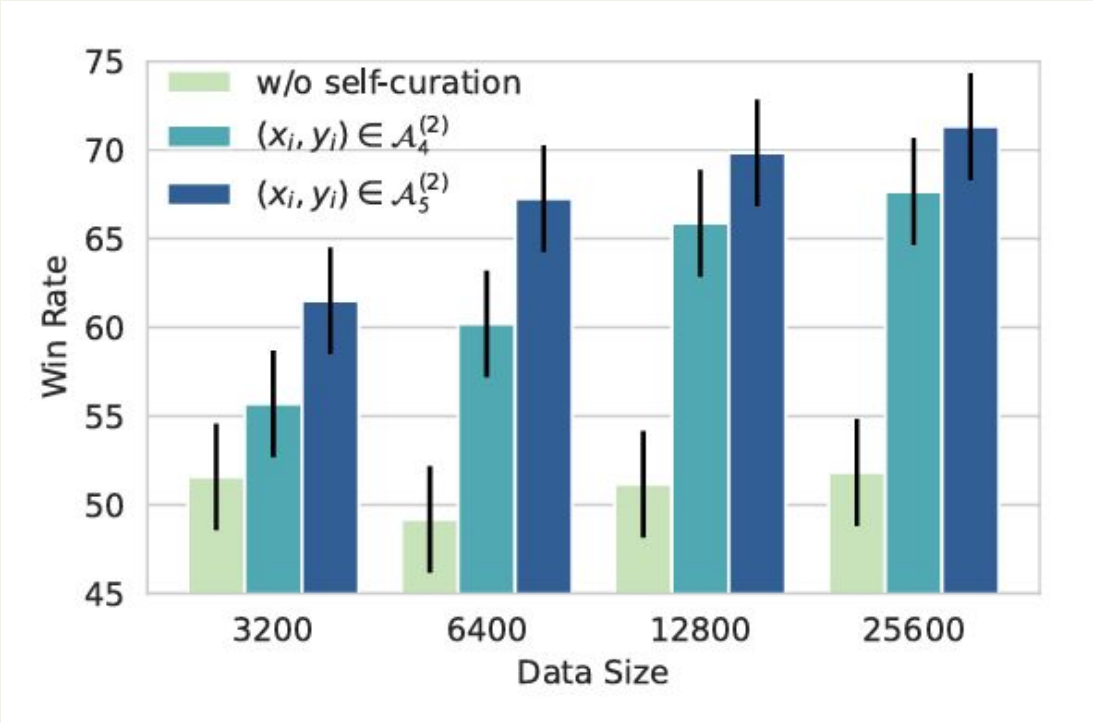


Experiments

Scaling Analysis: Data Quality

Data quality matters more than data quantity

- Training on large amounts of uncurated data does not improve performance

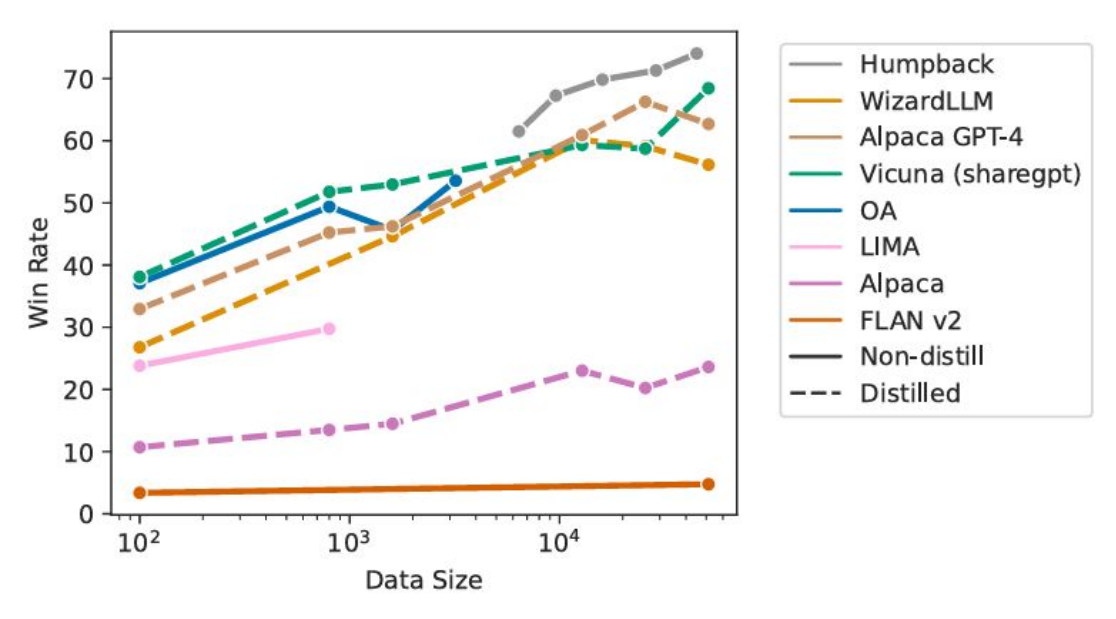


Experiments

Scaling Analysis: Data Efficiency

Instruction backtranslation achieves higher scaling efficiency than many existing instruction datasets

Distilled instruction datasets tend to have better data efficiency than datasets created from other sources, e.g. NLP tasks (FLAN v2) or extracted from community Q&A (LIMA)



Ablation Studies: Reasoning

❓ Evaluate whether instruction backtranslation improves the model's zero-shot reasoning abilities compared to the base LLaMA model

✅ Instruction backtranslation not only improves instruction-following but also strengthens reasoning capabilities across several benchmarks, especially more complex reasoning tasks.

Benchmarks used

- SIQA: social commonsense reasoning
- PIQA: physical commonsense reasoning
- ARC-E / ARC-C: grade-school science reasoning (easy / challenge)
- OpenBookQA: science knowledge reasoning
- MMLU: broad multi-task language understanding

	SIQA	PIQA	Arc-E	Arc-C	OBQA	MMLU
LLaMA 33B	50.2	82.2	80.0	54.8	58.6	49.5
Humpback 33B	53.4	74.5	84.4	68.5	46.4	55.4
LLaMA 65B	52.3	82.8	78.9	56.0	60.2	54.8
Humpback 65B	60.4	78.9	88.7	73.0	64.0	59.0

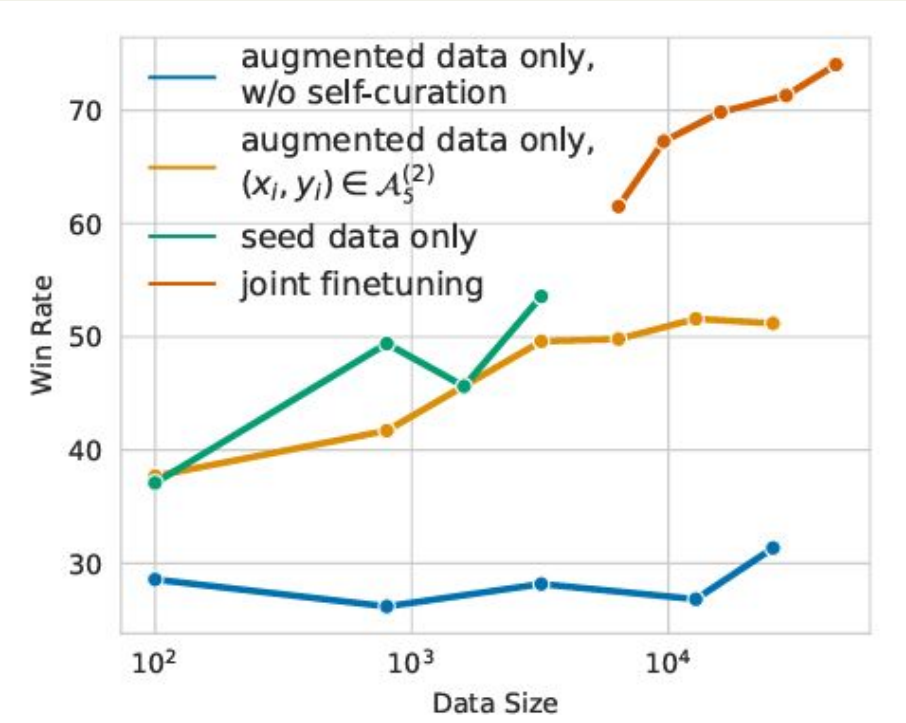
Experiments

Ablation Studies: Data Type

❓ Evaluate how self-augmented data and seed data contribute to model performance in instruction-following

✅Data quality is critical for synthetic training data (blue vs gold lines)

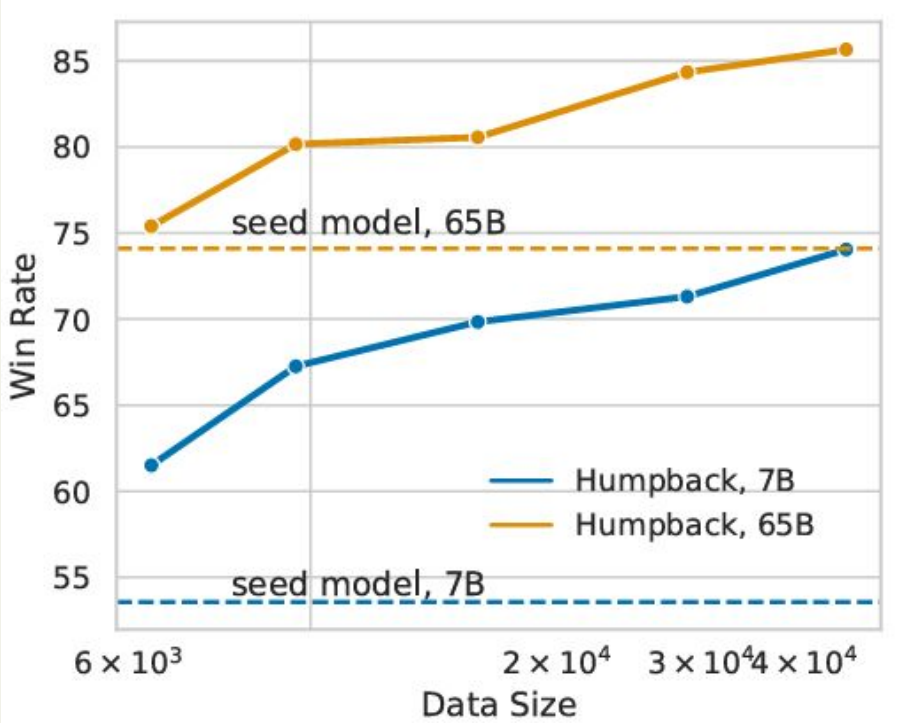
✅Seed data and augmented data are complementary: seed data provides correct assistant-style responses, while augmented data increases instruction diversity (orange line increase)



Ablation Studies: Data Amount

❓ Evaluate whether increasing the amount of self-curated A_5 instruction data continues to improve performance for both small and large models.

✅Both models performance continues to increase as A_5 training data grows, showing that models are not saturated even at ~40k instructions



Ablation Studies: Self-Curation

❓ Evaluate how well different models can select high quality instruction/response pairs during the self-curation step

✅As the model improves, it becomes better at identifying useful training data, which helps explain why the iterative self-curation process improves overall model quality

A dev set of 250 augmented examples was labeled by a human as high or low quality

Models are used to predict which examples are good training data

	Precision	Recall	Win Rate (%)
M_0	0.44	0.09	35.71 ±3.02
M_1	0.52	0.44	37.70 ±3.06
GPT-4	0.88	0.92	41.04±3.11

Ablation Studies: System Prompt

❓ Evaluate whether adding system prompts to distinguish seed data from augmented data affects model performance

✅Explicitly marking the source of data (seed vs augmented) while both training and at inference leads to best performance

Train	Inference	Win Rate (%)
S_a for seed data, S_w for augmented data	$\{S_a, S_w\}$	66.47 $\pm$3.04
no system prompt	no system prompt	59.96 $\pm$3.09
S_a for seed data, S_w for augmented data	S_a	62.69 $\pm$3.06
S_a for seed data, S_w for augmented data	no system prompt	62.70 $\pm$3.07

Broader Impacts + Importance

Introduced a new way to collect instruction-tuning data with minimal human work

- Instruction-following models can improve themselves using unlabeled web data, reducing dependence on expensive human annotation or distillation from proprietary models
- Scalable alignment method that generates large instruction datasets from small seed datasets
- Makes instruction tuning more accessible for research groups without access to large labeled datasets

Limitations & Future Work

Limitations:

- Synthetic training data generated by models may propagate errors
- Lack of Safety/Red-Teaming
- Method could not generate high quality responses for instructions which specify some specific formats, e.g. ASCII art

Future work can:

- Utilize larger corpora
- Incorporate safety limits
- Investigate generation for specific formats

Prompt specifying OOD content:
Make ASCII art of a cat
Can you make ASCII art? If so, can you show me a house?
Hi. I want to make an embossed picture for my blind girlfriend. There is a Braille printer at my university that I can use. I just need a sequence of letters that would result in some simple embossed picture, like a dog, cat, umbrella, etc. Can you give me a such text?
take the phone conversation in the movie Taken that Bryan Mills has with his daughters captors, and rewrite it in old english

LIMA: Less Is More for Alignment

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Avia Efrat[τ] Ping Yu[μ] Lili Yu[μ] Susan Zhang[μ] Gargi Ghosh[μ] Mike Lewis[μ] Luke Zettlemoyer[μ]
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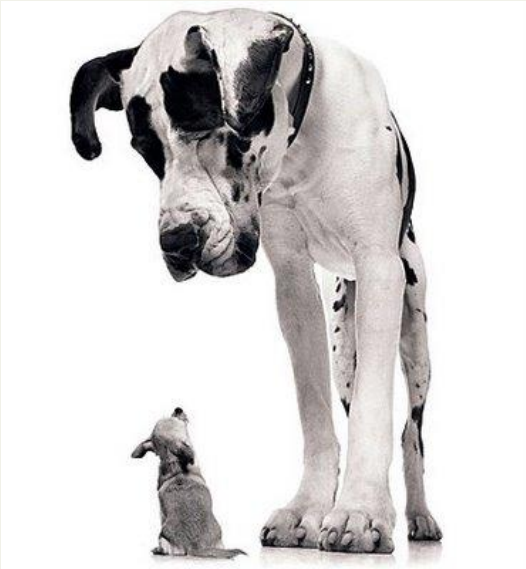
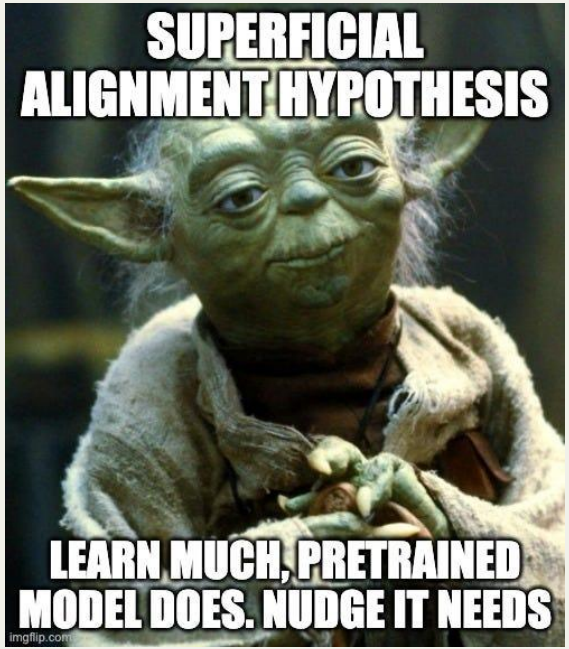
Introduction

- Modern LLMs are trained with large alignment datasets and complex RLHF pipelines
- Alignment typically relies on millions of instruction examples or RLHF
- These processes require extensive compute, data, and human labeling
- How much alignment data is actually necessary?



Motivation

- Authors propose the Superficial Alignment Hypothesis
- Most knowledge and reasoning ability is learned during pre training
- Alignment does not inject new knowledge, it teaches the model to select correct sub distribution from existing pre trained knowledge
- ‘Frozen’ Circuitry → actual reasoning & knowledge retrieval circuits remain largely fixed during alignment
    - Except output filter is reshaped
- If true, very little instruction data may be needed



Introduction

Overview

- Fine tunes a large pretrained language model using only 1000 High quality prompts
- LIMA is a 65B parameter LLaMA model
- Data set focuses on carefully curated and manually written examples rather than large automatically generated datasets
- Uses standard supervised fine tuning, without RLHF
- Designed to test whether small high quality datasets are enough

LLaMA

Alignment Data

- Training dataset contains 1,000 prompt response pairs → Supervised fine tuning
- Data focuses on high quality responses with clear reasoning and structure
- Prompts cover diverse categories including knowledge questions, advice, reasoning, and creative tasks
- Examples come from community forums and manual authoring
- Total dataset size: roughly 750k tokens

Source	#Examples	Avg Input Len.	Avg Output Len.
Training			
Stack Exchange (STEM)	200	117	523
Stack Exchange (Other)	200	119	530
wikiHow	200	12	1,811
Pushshift r/WritingPrompts	150	34	274
Natural Instructions	50	236	92
Paper Authors (Group A)	200	40	334
Dev			
Paper Authors (Group A)	50	36	N/A
Test			
Pushshift r/AskReddit	70	30	N/A
Paper Authors (Group B)	230	31	N/A

- StackExchange examples focus on technical and general knowledge questions
- wikiHow provides long instructional explanations, increasing output length diversity

Alignment Data

Community Questions and Answers

- 750 examples collected from online communities
- Sources include:
  - Stack Exchange
  - wikiHow
  - Reddit WritingPrompts
- Responses filtered for
  - Quality Self contained answers
  - Assistant like tone

Alignment Data

Manually Authored Examples

- 250 examples are written by the paper authors
- Prompts cover diverse real world scenarios
- Focus on complex prompts that require structured reasoning or explanation
- Responses follow a consistent assistant tone → clear + helpful + conversational
- Model learns desired response style and tone
- Includes safety related prompts requiring refusal

Training LIMA

- Start from LLaMA 65B pretrained model
- Fine tune using standard supervised loss
- Train for 15 epochs with AdamW optimizer
  - beta_1 = 0.9, beta_2 = 0.95, and weight decay of 0.1
  - Batch size = 32
  - < 2048 tokens
- Special end of turn token distinguishes speakers

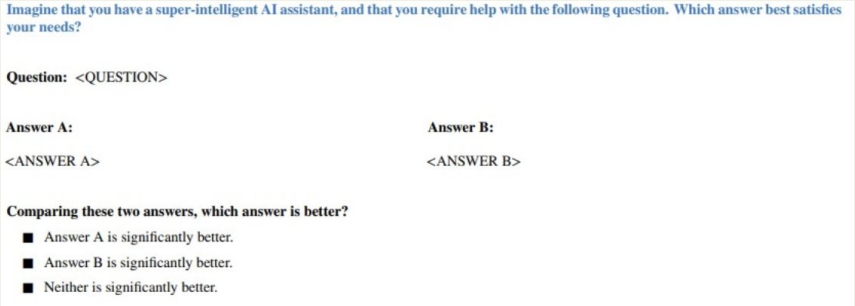
Human Evaluation

- Experiment Setup
- Results
- Analysis

Human Evaluation

Experiment Setup

- LIMA evaluated using human preference comparisons
- Compared against major models:
  - GPT-4
  - Claude
  - Bard
  - Alpaca



Imagine that you have a super-intelligent AI assistant, and that you require help with the following question. Which answer best satisfies your needs?

Question: <QUESTION>

Answer A:

<ANSWER A>

Answer B:

<ANSWER B>

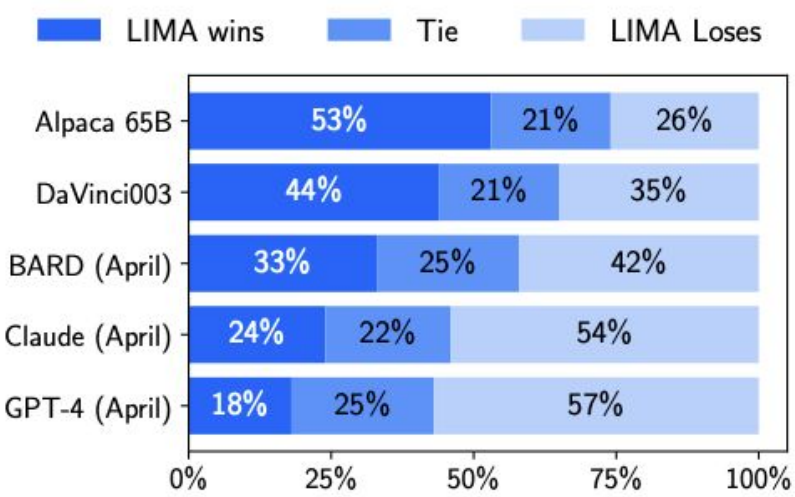
Comparing these two answers, which answer is better?

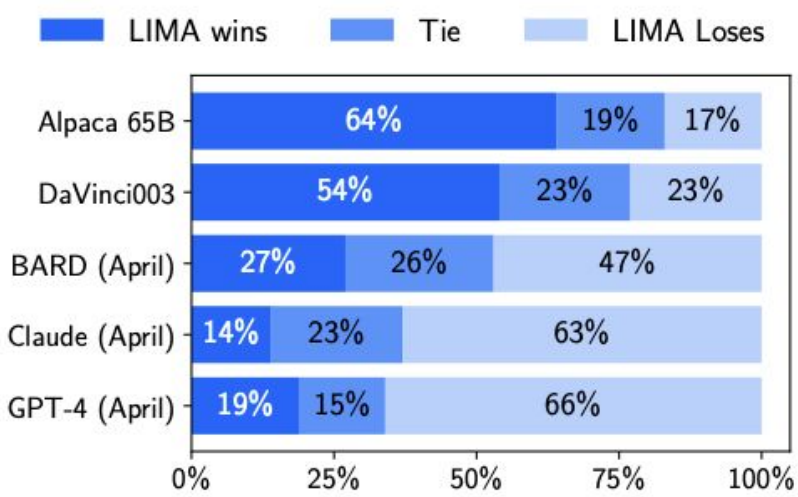
- Answer A is significantly better.
- Answer B is significantly better.
- Neither is significantly better.

- Prompts = various real world instruction tasks
- Human annotators are shown 2 responses 2 the same prompt
- Annotators select which response is better

Human Evaluation

Results





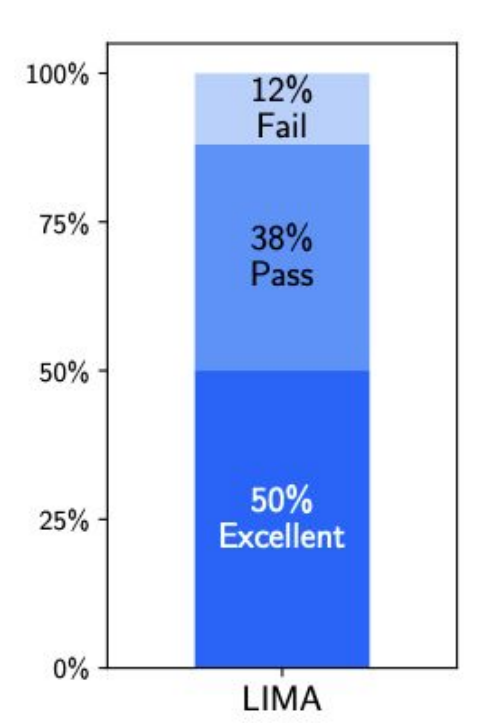
- LIMA beats Alpaca 65B despite using far less data
- LIMA often outperforms DaVinci003 with RLHF

- Compared to GPT-4:
  - LIMA responses preferred 19% of the time
- Shows strong performance with little alignment data

Human Evaluation

Results

- LIMA responses across 500 Eval Prompts →
- Most = highly rated for clarity, reasoning, and helpfulness
- Strong performance appears in knowledge questions, explanations, and structured answers
- Lower scores often occur in tasks requiring deeper reasoning or nuanced judgment
- Suggests LIMA learns response style effectively even with limited training data



Human Evaluation

Analysis

- LIMA outperforms several strong models in human preference comparisons
- Only 1,000 training examples, responses remain competitive with large aligned systems
- Performance suggests most knowledge comes from pretraining, not alignment data scale
- Small high quality datasets can effectively teach style, structure, and helpfulness
- LIMA still loses frequently to GPT-4 → limits of minimal alignment.

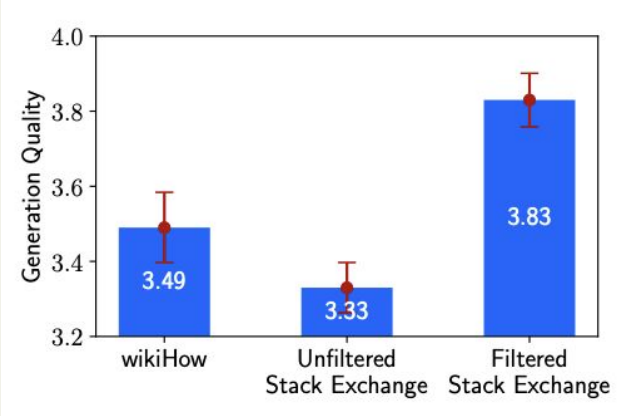
Style vs. Reasoning: Alignment Limit

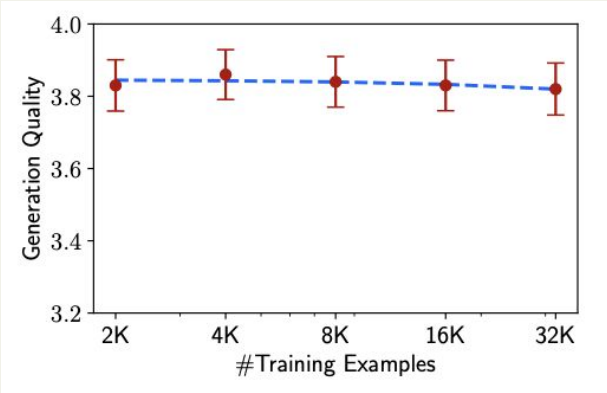
- Alignment is about learning style to interact with users → effectively exposing capabilities
- While 1,000 examples are enough to teach the tone of a helpful assistant, they do not seem to reconfigure the model's fundamental logic
- Performance Gap:
  - Success: High ratings for clarity, structure, and following complex formatting.
  - Failure: Struggles with tasks requiring nuanced judgment or deeper reasoning than what was acquired during pre training.
- Key Takeaway: Alignment teaches the model how to speak, not how to think
  - If a capability wasn't learned during pre-training, alignment is unlikely to inject it

Ablations on Data Diversity, Quality, and Quantity

Why is Less More?

- Prompt diversity improves performance significantly
- Higher response quality also improves results
- Performance improves rapidly with the first few hundred
- Simply adding more examples shows diminishing returns - (around 1k examples)
- Teaches response format + behavior than knowledge
- Alignment depends more on diversity and quality than scale





Multiturn Dialogue

Averaged over 10 conversations

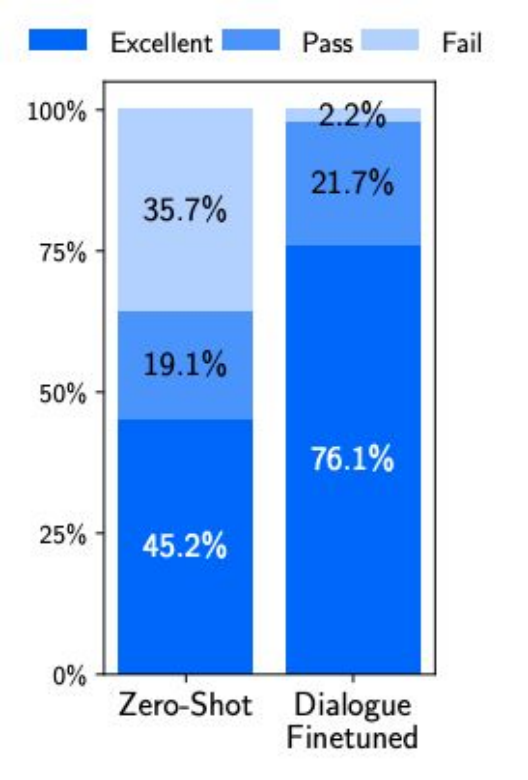
LIMA was only fine tuned on single-turn interactions, can it also perform on multi-turn dialogue?

Excellent, Pass, Fail scoring

Original LIMA → No (zero-shot)

New LIMA → Yes

- Fine tuned on 30 additional multi-turn examples
- Failure rate dropped from 15/42 turns to 1/42 turns
- “Excellent”s increase 45.2% → 76.1%



Broader Impacts + Importance

Showed that fine-tuning a LM on a small set of carefully curated examples can produce competitive results on a wide range of prompts.

Superficial Alignment Hypothesis → introduced core idea that a model's capabilities are learned from pretraining, and fine tuning is about learning style.

Presented additional evidence that simple extensions to data can increase capabilities substantially while the set remains small (multi-turn dialogue)

Limitations

Scalability

Significant effort goes into constructing the data and difficult to scale up

- High quality curation requires export manual labor to ensure diversity and stylistic consistency → harder to automate than synthetic generation

Robustness

An unlucky sample during decoding or an adverse prompt can weaken the response

- Without safety guardrails, the model is more sensitive to specific prompt phrasing

Evaluation

Only a very small test set

- 300 test prompts and human preference labels make it difficult to statically capture the model's full tail of edge cases or rare knowledge failures

Recap

Paper	Motivation	Solution	Relation to Others
Self-Instruct	Find a cheaper way to build large instruction tuning dataset	Bootstrap a small seed dataset into a large, quality dataset	Provides a method to generate large, high quality instruction sets.
Self-Alignment	Find a cheaper way to build large instruction tuning dataset	Treat raw data as output, generate the related input	Alternative to Self-Instruct, backtranslation vs. bootstrapping
LIMA	Show instruction tuning can be done without large datasets	Create a small, high quality dataset that rivals larger ones	Shows large dataset collection is unnecessary

Final Thoughts

Instruction tuning is a core component of modern LMs, enabling them to generalize across diverse tasks and interact naturally with users.

The traditional approach of using large-scale human-written instruction data is expensive to collect and difficult to ensure quality.

These papers move away from the assumption that large, human-annotated datasets are necessary, and toward alternative methods that decrease the cost and complexity of instruction tuning, whether through automatic data generation or careful quality-driven curation.

Thank you!

Questions?